

Aman Choudhary

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PROFESSIONAL SUMMARY

Software Development Engineer with 3 years of experience at **Philips Healthcare**, building scalable backend services for medical device fleet management. Proficient in **Java/Spring Boot microservices**, **event-driven architecture (Kafka)**, and cloud-native deployments on **AWS (EKS, RDS, ElastiCache, S3)**. Experienced in designing distributed systems for **IoT platforms** managing 100+ MRI scanners across hospitals with **HIPAA-compliant** data handling.

PROFESSIONAL EXPERIENCE

Philips Healthcare

May 2024 – Present

Software Development Engineer

Bangalore, India

- Part of a backend team building a **cloud-based fleet management platform** for centralized monitoring and configuration management of 100+ MRI scanners deployed across hospitals.
- Owned **Config Transfer Service** — designed configuration export/import pipeline using **Strategy Pattern** to handle 8+ file types (SIS, Shim, PSC), reducing field engineer intervention by 60% for software upgrades.
- Built **Fleet Health Service** with real-time heartbeat monitoring via **WebSocket (STOMP)**; implemented a **Redis TTL-based offline detection** pattern — key expiry triggers machine-offline alerts without polling.
- Designed event-driven architecture using **Apache Kafka** for async config transfer events, heartbeat aggregation, and audit trail publishing with guaranteed delivery using **idempotent producers** and **Dead Letter Topics**.
- Implemented **Redis caching** for session validation (TTL=30min, avoiding Auth service calls per request) and machine metadata (TTL=5min), reducing average API response time from 120ms to 15ms.
- Designed database schema with **Flyway migrations**, used **JPA Specifications** for dynamic query building (filter by hospital, status, date range), and **optimistic locking** for concurrent config updates.
- Implemented **Spring Security + JWT** with role-based access (ADMIN, ENGINEER, VIEWER), **audit logging** for all configuration changes (HIPAA compliance), and secure file storage on **AWS S3**.
- Containerized services with **Docker** (multi-stage builds, non-root user), deployed on **AWS EKS** with **Helm charts**, **HPA** auto-scaling, and **GitHub Actions** CI/CD (build → SonarQube → deploy).
- Set up observability with **Micrometer + Prometheus + Grafana** dashboards, **structured JSON logging** with correlation IDs, and custom alerts for heartbeat failures and transfer errors.

Philips Healthcare

Jun 2023 – Apr 2024

Software Development Engineer Intern

Bangalore, India

- Developed automated installation and deployment framework for MRI scanner software using **Java**, reducing manual setup time by 70% across 50+ hospital deployments.
- Built validation modules with **JUnit** and **Mockito**, achieving 85%+ code coverage; identified and fixed 12 critical deployment edge cases through systematic testing.
- Implemented multi-threaded parallel processing for deployment tasks using **ExecutorService**, improving deployment pipeline throughput by 50%.
- Wrote SQL scripts for configuration data validation and migration, collaborated with QA on integration testing across multiple scanner software versions.

TECHNICAL SKILLS

Languages: Java, SQL

Backend: Spring Boot, Spring MVC, Spring Data JPA, Spring Security (JWT/OAuth2), Hibernate, MapStruct, WebSocket (STOMP)

Databases & Caching: PostgreSQL, Redis (ElastiCache), Flyway Migrations

Messaging & Patterns: Apache Kafka, Strategy Pattern, Event-Driven Architecture, Saga (Choreography), Builder

Cloud & DevOps: AWS (EKS, RDS, ElastiCache, S3, CloudWatch, IAM), Docker, Kubernetes, Helm, GitHub Actions, SonarQube

Observability & Tools: Micrometer, Prometheus, Grafana, Structured Logging (JSON), Maven, Git, Swagger/OpenAPI, Postman

AI & Productivity: GitHub Copilot, AI-assisted code generation & review, prompt engineering for development workflows

EDUCATION

Lovely Professional University

Aug 2020 – May 2024

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.08/10